Arrival · Supported

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Pair

What does heterozygous Aa mean?

Help: Read or point first. Compare the two alleles.

2. Gamete

How many alleles for this gene go in one gamete?

Help: Read or point first. One parent contributes one to the offspring pair.

3. Options

Which gamete alleles can an Aa parent produce?

Help: Read or point first. Separate the pair, not copy Aa into every margin.

4. Combine

A gamete with A combines with a gamete with a. What is the offspring genotype?

Help: Read or point first. Use one allele from each.

My response: _____

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Independent · Supported

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Construct

Complete Aa × Aa with labelled gametes and four offspring boxes.

Help: Write parents first. Put one allele in each margin. Fill each box from its column and row.

2. Compare

Construct Aa × aa and state the chance of aa.

Help: Count matching boxes ___ out of four. Divide by four and multiply by 100 to give the percentage.

3. Explain

Why do two boxes sometimes contain the same genotype?

Help: Trace the column and row for each matching box. Different gamete combinations can give the same allele pair.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Supported

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

1. Construct

For Aa × Aa, list the four predicted genotype boxes.

Help: Use: my answer is ___ because ___. Point to a model or card first. Combine the row and column labels.

2. Interpret

What fraction of boxes is heterozygous?

Help: Use: my answer is ___ because ___. Point to a model or card first. Count Aa, including the equivalent aA combination.

My response: _____

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Build the cross, then check the chance

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Complete the four boxes for Aa × Aa. Use one allele from each margin. Then change the second parent to aa and compare the two crosses.

Cross	Parent 1 gametes	Parent 2 gametes
Aa × Aa	A or a	A or a
Aa × aa	A or a	a only

My evidence and reasoning

One limit or useful check

My response: _____

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.

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Punnett square · working sheet

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Record the parent alleles. Fill all four boxes. Then find the chance of aa.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Compare with Aa × aa.

Aa × aa · side / top	**A**	**a**
a	_____	_____
a	_____	_____

Aa × Aa chance of aa: ____ . Aa × aa chance of aa: ____ .

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